

Introduction to Machine Learning
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Lecturer: Zhihui Li, Xiaojun Chang
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Name: Jinghao Chen

Homework 1
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ID: PB23010359

Notice, to get the full credits, please present your solutions step by step.

Exercise 1: Limit and Limit Points

1. Show that $\{\mathbf{x}_n\}$ in $\mathbb{R}^n$ converges to $\mathbf{x} \in \mathbb{R}^n$ if and only if $\{\mathbf{x}_n\}$ is bounded and has a unique limit point $\mathbf{x}$.
2. (**Limit Points of a Set**). Let C be a subset of $\mathbb{R}^n$. A point $\mathbf{x} \in \mathbb{R}^n$ is called a limit point of C if there is a sequence $\{\mathbf{x}_n\}$ in C such that $\mathbf{x}_n \rightarrow \mathbf{x}$ and $\mathbf{x}_n \neq \mathbf{x}$ for all positive integers n . If $\mathbf{x} \in C$ and $\mathbf{x}$ is not a limit point of C , then $\mathbf{x}$ is called an isolated point of C . Let C' be the set of limit points of the set C . Please show the following statements.
 - (a) If $C = (0, 1) \cup \{2\} \subset \mathbb{R}$, then $C' = [0, 1]$ and $x = 2$ is an isolated point of C .
 - (b) The set C' is closed.

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Solution 1: Limit and Limit Points

1. ①Necessity:

$$\lim_{n \rightarrow \infty} x_n = x \implies \forall \epsilon > 0, \exists N > 0, \text{ s.t. when } n \geq N, \|x_n - x\| < \epsilon \implies \|x_n\| < \|x\| + \epsilon.$$

Set

$$M = \max\{\|x_1\|, \|x_2\|, \dots, \|x_{N-1}\|, \|x\| + \epsilon\}.$$

Clearly, $\|x_n\| \leq M, \forall n > 0$, i.e., sequence $\{x_n\}$ is bounded.

Besides, $\lim_{n \rightarrow \infty} x_n = x \implies$ all subsequences $\{x_{n_k}\}$ also converge to $x \implies$ limit point x is unique.

②Sufficiency:

Suppose, for the sake of contradiction, that $\lim_{n \rightarrow \infty} x_n \neq x$. Then $\exists \epsilon_0 > 0$ and a subsequence $\{x_{n_k}\}$ such that

$$\|x_{n_k} - x\| \geq \epsilon_0, \forall k > 0$$

According to Bolzano-Weierstrass Theorem, $\{x_{n_k}\}$ is bounded $\implies$ there is a further convergent subsequence $\{x_{n_{k_j}}\}$ with $\lim_{j \rightarrow \infty} x_{n_{k_j}} = y \in \mathbb{R}^n$. By the uniqueness of limit points, $y = x$. Therefore, $\|x_{n_{k_j}} - x\| \rightarrow 0$, which contradicts $\|x_{n_k} - x\| \geq \epsilon_0, \forall k > 0$.

Hence our supposition was false and $x_n \rightarrow x$.

2. (a) For any $x \in [0, 1]$, take $x_n = x + \frac{1}{n}$ if $x < 1$; $x_n = x - \frac{1}{n}$ if $x > 0$. Then it is clear that $x_n \neq x, \forall n > 0$ and $x_n \rightarrow x$. Thus, x is a limit point of C .

If $x \notin [0, 1]$, $\text{dist}(x, (0, 1)) > 0 \implies \exists \epsilon > 0$, s.t. $B(x, \epsilon) \setminus \{x\} \cap C = \emptyset \implies x$ is not a limit point of C .

Thus $C' = [0, 1]$. Correspondingly, $x = 2$ is an isolated point of C .

- (b) To prove C' is closed $\iff$ to prove $C' = \overline{C'} \iff$ to prove all of the limit points of C' belong to C'

Set x as any limit point of $C' \implies$ there exists a sequence $\{x_n\} \subset C' \setminus \{x\}$ with $x_n \rightarrow x$.

For each n , since $x_n \in C'$, there exists a sequence $\{x_{n,m}\} \subset C' \setminus \{x_n\}$ with $x_{n,m} \rightarrow x_n$.

$\forall \epsilon > 0, \exists N > 0$, s.t. when $n \geq N, \|x_n - x\| < \frac{\epsilon}{2}$. Let m be large enough, then $\|x_{n,m} - x_n\| < \frac{\epsilon}{2}$. By triangle inequality,

$$\|x_{n,m} - x\| \leq \|x_{n,m} - x_n\| + \|x_n - x\| \leq \epsilon$$

Therefore, $x_{n,m} \in C$ implies $x \in C'$.

In conclusion, the set C' is closed.

■

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Exercise 2: Norms

In this exercise, we will give some examples of norms and a useful theorem related to norms in **finite** dimensional vector space.

1. **l_p norm:** The l_p norm is defined by

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$$

where $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ and $p \geq 1$.

- (a) Please show that the l_p norm is a norm.
- (b) Please show that the following equality.

$$\lim_{p \rightarrow \infty} \|\mathbf{x}\|_p = \|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

The l_∞ norm is defined as above.

2. **Operator norms:** Suppose that $\mathbf{A} \in \mathbb{R}^{m \times n}$, which can be viewed as a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$. Please show the following operator norms' equality.

- (a) Let $\|\mathbf{A}\|_1 = \sup_{\mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{Ax}\|_1}{\|\mathbf{x}\|_1}$. Please show that

$$\|\mathbf{A}\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}|.$$

- (b) Let $\|\mathbf{A}\|_\infty = \sup_{\mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{Ax}\|_\infty}{\|\mathbf{x}\|_\infty}$. Please show that

$$\|\mathbf{A}\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}|.$$

3. **(Optional) Dual norm:** Let $\|\cdot\|$ be a norm on $\mathbb{R}^n$. The dual norm of $\|\cdot\|$ is defined by

$$\|\mathbf{x}\|_* = \sup_{\mathbf{y} \in \mathbb{R}^n, \|\mathbf{y}\| \leq 1} \mathbf{y}^\top \mathbf{x}.$$

- (a) Please show that the dual of the Euclidean norm is the Euclidean norm itself. i.e.,

$$\sup_{\mathbf{y} \in \mathbb{R}^n, \|\mathbf{y}\|_2 \leq 1} \mathbf{y}^\top \mathbf{x} = \|\mathbf{x}\|_2.$$

- (b) Please show that the dual of the l_1 norm is the l_∞ norm. i.e.,

$$\sup_{\mathbf{y} \in \mathbb{R}^n, \|\mathbf{y}\|_1 \leq 1} \mathbf{y}^\top \mathbf{x} = \|\mathbf{x}\|_\infty.$$

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Solution 2: Norms

1. (a) ① Positive Definiteness:

$$|x_i| \geq 0 \implies \|\mathbf{x}\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p} \geq 0$$

where the equality holds if and only if $x_i = 0, \forall 1 \leq i \leq n \iff \mathbf{x} = \mathbf{0}$.

② Homogeneity: $\forall \alpha \in \mathbb{R}$,

$$\|\alpha \mathbf{x}\|_p = \left(\sum_{i=1}^n |\alpha x_i|^p \right)^{1/p} = |\alpha| \left(\sum_{i=1}^n |x_i|^p \right)^{1/p} = |\alpha| \|\mathbf{x}\|_p$$

③ Triangle Inequality: Let $q = \frac{p}{p-1}$ to satisfy $\frac{1}{p} + \frac{1}{q} = 1$.

$\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, according to Hölder inequality,

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\|_p^p &= \sum_{i=1}^n |x_i + y_i|^p \\ &\leq \sum_{i=1}^n |x_i| \cdot |x_i + y_i|^{p-1} + \sum_{i=1}^n |y_i| \cdot |x_i + y_i|^{p-1} \\ &\leq \left(\sum_{i=1}^n |x_i|^p \right)^{1/p} \cdot \left(\sum_{i=1}^n |x_i + y_i|^{(p-1)q} \right)^{1/q} \\ &\quad + \left(\sum_{i=1}^n |y_i|^p \right)^{1/p} \cdot \left(\sum_{i=1}^n |x_i + y_i|^{(p-1)q} \right)^{1/q} \\ &= (\|\mathbf{x}\|_p + \|\mathbf{y}\|_p) \cdot \left(\sum_{i=1}^n |x_i + y_i|^p \right)^{1/q} \\ \implies \|\mathbf{x} + \mathbf{y}\|_p &= \left(\sum_{i=1}^n |x_i + y_i|^p \right)^{1/p} = \left(\sum_{i=1}^n |x_i + y_i|^p \right)^{1-1/q} \leq \|\mathbf{x}\|_p + \|\mathbf{y}\|_p \end{aligned}$$

- (b) It is clear that

$$\|\mathbf{x}\|_\infty^p \leq \|\mathbf{x}\|_p^p \leq n \|\mathbf{x}\|_\infty^p \iff \|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_p \leq n^{1/p} \|\mathbf{x}\|_\infty$$

By the Squeeze Theorem,

$$\lim_{p \rightarrow \infty} n^{1/p} = 1 \implies \lim_{p \rightarrow \infty} \|\mathbf{x}\|_p = \|\mathbf{x}\|_\infty$$

2. When $\mathbf{A} = \mathbf{0}$, the conclusion is trivial. Thus, the following discussion is based on $\mathbf{A} \neq \mathbf{0}$.

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Divide matrix $\mathbf{A}$ into blocks by columns as $(a_1, a_2, \dots, a_n)$ and let $\|a_{j_0}\|_1 = \max_{1 \leq j \leq n} \|a_j\|_1$.

Then $\forall \mathbf{x} \in \mathbb{R}^n$ that satisfies $\|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i| = 1$,

$$\|\mathbf{Ax}\|_1 = \left\| \sum_{j=1}^n x_j a_j \right\|_1 \leq \sum_{j=1}^n |x_j| \|a_j\|_1 \leq \left(\sum_{i=1}^n |x_i| \right) \max_{1 \leq j \leq n} \|a_j\|_1 = \|a_{j_0}\|_1$$

Besides, $\|\mathbf{Ae}_{j_0}\|_1 = \|a_{j_0}\|_1$. In conclusion,

$$\|\mathbf{A}\|_1 = \max_{\|\mathbf{x}\|_1=1} \|\mathbf{Ax}\|_1 = \max_{1 \leq j \leq n} \|a_j\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}|.$$

(b) $\forall \mathbf{x} \in \mathbb{R}^n$ that satisfies $\|\mathbf{x}\|_\infty = 1$,

$$\|\mathbf{Ax}\|_\infty = \max_{1 \leq i \leq n} \left| \sum_{j=1}^n a_{ij} x_j \right| \leq \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}| |x_j| \leq \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}|$$

Set $\max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}| = \sum_{j=1}^n |a_{kj}|$. Then let $\tilde{\mathbf{x}} = (\text{sgn}(a_{k1}), \dots, \text{sgn}(a_{kn}))^T$.

$\mathbf{A} \neq \mathbf{0} \implies \|\tilde{\mathbf{x}}\|_\infty = 1$ and it is clear that $\|\mathbf{A}\tilde{\mathbf{x}}\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}|$.

3. (a) By Cauchy-Schwarz Inequality,

$$\mathbf{y}^T \mathbf{x} \leq \|\mathbf{y}\|_2 \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_2 \implies \sup_{\mathbf{y} \in \mathbb{R}^n, \|\mathbf{y}\|_2 \leq 1} \mathbf{y}^T \mathbf{x} \leq \|\mathbf{x}\|_2.$$

If $\mathbf{x} = \mathbf{0}$, the target equality is trivial; if not, choose $\mathbf{y}_0 = \frac{\mathbf{x}}{\|\mathbf{x}\|_2}$, which satisfies $\|\mathbf{y}_0\|_2 = 1$. Then $\mathbf{y}_0^T \mathbf{x} = \frac{\mathbf{x}^T \mathbf{x}}{\|\mathbf{x}\|_2} = \|\mathbf{x}\|_2$.

In conclusion, $\sup_{\mathbf{y} \in \mathbb{R}^n, \|\mathbf{y}\|_2 \leq 1} \mathbf{y}^T \mathbf{x} = \|\mathbf{x}\|_2$.

(b) For any $\mathbf{y} = (y_1, \dots, y_n)^T$ with $\|\mathbf{y}\|_1 \leq 1$,

$$\begin{aligned} \mathbf{y}^T \mathbf{x} &= \sum_{i=1}^n x_i y_i \leq \sum_{i=1}^n |x_i| |y_i| \leq \max_{1 \leq i \leq n} |x_i| \sum_{i=1}^n |y_i| \leq \|\mathbf{x}\|_\infty \\ &\implies \sup_{\mathbf{y} \in \mathbb{R}^n, \|\mathbf{y}\|_1 \leq 1} \mathbf{y}^T \mathbf{x} \leq \|\mathbf{x}\|_\infty. \end{aligned}$$

If $\mathbf{x} = \mathbf{0}$, the target equality is trivial; if not, let $k = \arg \max_{1 \leq i \leq n} |x_i|$ and choose $\mathbf{y}_0 = \text{sgn}(x_k) \mathbf{e}_k$ so that $\|\mathbf{y}_0\|_1 = 1$. Then $\mathbf{y}_0^T \mathbf{x} = \text{sgn}(x_k) x_k = \|x_k\|_\infty$.

In conclusion, $\sup_{\mathbf{y} \in \mathbb{R}^n, \|\mathbf{y}\|_1 \leq 1} \mathbf{y}^T \mathbf{x} = \|\mathbf{x}\|_\infty$. ■

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Exercise 3: Open and Closed Sets

The norm ball $\{\mathbf{y} \in \mathbb{R}^n : \|\mathbf{y} - \mathbf{x}\|_2 < r, \mathbf{x} \in \mathbb{R}^n\}$ is denoted by $B_r(\mathbf{x})$.

1. Given a set $C \subset \mathbb{R}^n$, please show the following are equivalent.

- (a) The set C is closed; that is $\mathbf{cl} C = C$.
- (b) The complement of C is open.
- (c) If $B_\epsilon(\mathbf{x}) \cap C \neq \emptyset$ for every $\epsilon > 0$, then $\mathbf{x} \in C$.

2. Given $A \subset \mathbb{R}^n$, a set $C \subset A$ is called open in A if

$$C = \{\mathbf{x} \in C : B_\epsilon(\mathbf{x}) \cap A \subset C \text{ for some } \epsilon > 0\}.$$

A set C is said to be closed in A if $A \setminus C$ is open in A .

- (a) Let $B = [0, 1] \cup \{2\}$. Please show that $[0, 1]$ is not an open set in $\mathbb{R}$, while it is both open and closed in B .
- (b) Please show that a set $C \subset A$ is open in A if and only if $C = A \cap U$, where U is open in $\mathbb{R}^n$.

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Solution 3: Open and Closed Sets

1. (a) $\implies$ (b):

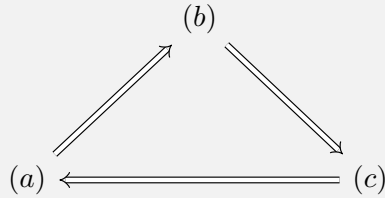
Suppose, for the sake of contradiction, that C^c is not open. Then $\exists \mathbf{x} \in C$, s.t. $\forall \epsilon > 0, B_\epsilon(\mathbf{x}) \not\subset C^c \implies \text{take } \mathbf{x}_n \in B_\epsilon(\mathbf{x}) \setminus C^c \subset C \text{ for } \epsilon_n > 0$. Without generality, set $\epsilon_1 > \epsilon_2 > \dots > \epsilon_n \rightarrow 0$, then it is clear that sequence $\{\mathbf{x}_n\}$ satisfies $\lim_{n \rightarrow \infty} \mathbf{x}_n = \mathbf{x}$. Therefore, $\mathbf{x} \in \text{cl } C = C$, which contradicts $\mathbf{x} \in C^c$.

(b) $\implies$ (c):

Suppose, for the sake of contradiction, that $\exists \mathbf{x} \in C^c$, s.t. $B_\epsilon(\mathbf{x}) \cap C \neq \emptyset$ for every $\epsilon > 0$. Because C^c is open, $\exists \delta > 0$, s.t. $B_\delta(\mathbf{x}) \subset C^c \implies B_\delta(\mathbf{x}) \cap C = \emptyset$. So this is a contradiction.

(c) $\implies$ (a):

Suppose, for the sake of contradiction, that $\exists$ a sequence $\{\mathbf{x}_n\} \subset C$, which satisfies $\mathbf{x}_n \rightarrow \mathbf{x} \in C^c$, i.e., $\forall \epsilon > 0, \exists N > 0$, s.t. when $n \geq N, \|\mathbf{x}_n - \mathbf{x}\| < \epsilon$. Therefore, $B_\epsilon(\mathbf{x}) \cap C \supset \{\mathbf{x}_n\} \neq \emptyset \implies \mathbf{x} \in C$, which contradicts $\mathbf{x} \in C^c$.



In summary, these three hypothesis are equivalent.

2. (a) $\forall \epsilon > 0, -\frac{\epsilon}{2} \notin [0, 1] \implies B_\epsilon(0) = (-\epsilon, \epsilon) \not\subset [0, 1] \implies [0, 1]$ is not an open set in $\mathbb{R}$.

If $\mathbf{x} \in (0, 1)$, take $\epsilon = \min\{\mathbf{x}, 1 - \mathbf{x}\} > 0$. Then $B_\epsilon(\mathbf{x}) \subset [0, 1] \implies B_\epsilon(\mathbf{x}) \cap B = B_\epsilon(\mathbf{x}) \subset [0, 1]$.

If $\mathbf{x} = 0$, take $\epsilon = \frac{1}{2} > 0$. Then $B_\epsilon(0) = (-\frac{1}{2}, \frac{1}{2}) \implies B_\epsilon(0) \cap B = [0, \frac{1}{2}] \subset [0, 1]$. In the same way, if $\mathbf{x} = 1$, take $\epsilon = \frac{1}{2} > 0$. Then $B_\epsilon(1) = (\frac{1}{2}, \frac{3}{2}) \implies B_\epsilon(1) \cap B = [\frac{1}{2}, 1] \subset [0, 1]$.

In conclusion, $[0, 1]$ is open in B .

Beside, if $\mathbf{x} = 2$, take $\epsilon = \frac{1}{2} > 0$. Then $B_\epsilon(2) = (\frac{3}{2}, \frac{5}{2}) \implies B_\epsilon(2) \cap B = \{\frac{1}{2}\} \subset \{\frac{1}{2}\} \implies \{2\} = B \setminus [0, 1]$ is open in B .

In summary, $[0, 1]$ is both open and closed in B .

(b) ① Necessity:

$C \subset A$ is open in $A \implies \forall \mathbf{x} \in C, \exists \epsilon_x > 0$, s.t. $B_{\epsilon_x}(\mathbf{x}) \cap A \subset C \implies$

$$\bigcup_{\mathbf{x} \in C} (B_{\epsilon_x}(\mathbf{x}) \cap A) = \left(\bigcup_{\mathbf{x} \in C} B_{\epsilon_x}(\mathbf{x}) \right) \cap A \subset C.$$

Set $U = \bigcup_{\mathbf{x} \in C} B_{\epsilon_x}(\mathbf{x})$. Then $U \cap A \subset C$ and it is clear that U is open in $\mathbb{R}^n$.

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$\forall \mathbf{x} \in C, \mathbf{x} \in B_{\epsilon_x}(\mathbf{x}) \text{ and } \mathbf{x} \in C \subset A \implies$

$$\mathbf{x} \in B_{\epsilon_x}(\mathbf{x}) \cap A \subset \bigcup_{\mathbf{x} \in C} (B_{\epsilon_x}(\mathbf{x}) \cap A) = \left(\bigcup_{\mathbf{x} \in C} B_{\epsilon_x}(\mathbf{x}) \right) \cap A = U \cap A.$$

$\implies C \subset U \cap A.$

In conclusion, $C = U \cap A$ and U is open in $\mathbb{R}^n$.

② Sufficiency:

U is open in $\mathbb{R}^n \implies \forall \mathbf{x} \in U \supset C, \exists \epsilon_x > 0, \text{ s.t. } B_{\epsilon_x}(\mathbf{x}) \subset U \implies B_{\epsilon_x}(\mathbf{x}) \cap A \subset U \cap A = C.$ Thus $C \subset A$ is open in A .

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Exercise 4: Projection

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{x} \in \mathbb{R}^m$. Define

$$\mathbf{P}_{\mathbf{A}}(\mathbf{x}) = \underset{\mathbf{z} \in \mathbb{R}^m}{\operatorname{argmin}} \{ \|\mathbf{x} - \mathbf{z}\|_2 : \mathbf{z} \in \mathcal{C}(\mathbf{A}) \}.$$

We call $\mathbf{P}_{\mathbf{A}}(\mathbf{x})$ the projection of the point $\mathbf{x}$ onto the column space of $\mathbf{A}$.

1. Please show that $\mathbf{P}_{\mathbf{A}}(\mathbf{x})$ is unique for any $\mathbf{x} \in \mathbb{R}^m$.
2. Let $\mathbf{v}_i \in \mathbb{R}^n$, $i = 1, \dots, d$ with $d \leq n$, which are linearly independent.
 - (a) For any $\mathbf{w} \in \mathbb{R}^n$, please find $\mathbf{P}_{\mathbf{v}_1}(\mathbf{w})$, which is the projection of $\mathbf{w}$ onto the subspace spanned by $\mathbf{v}_1$.
 - (b) Please show $\mathbf{P}_{\mathbf{v}_1}(\cdot)$ is a linear map, i.e.,

$$\mathbf{P}_{\mathbf{v}_1}(\alpha \mathbf{u} + \beta \mathbf{w}) = \alpha \mathbf{P}_{\mathbf{v}_1}(\mathbf{u}) + \beta \mathbf{P}_{\mathbf{v}_1}(\mathbf{w}),$$

where $\alpha, \beta \in \mathbb{R}$ and $\mathbf{w} \in \mathbb{R}^n$.

- (c) Please find the projection matrix corresponding to the linear map $\mathbf{P}_{\mathbf{v}_1}(\cdot)$, i.e., find the matrix $\mathbf{H}_1 \in \mathbb{R}^{n \times n}$ such that

$$\mathbf{P}_{\mathbf{v}_1}(\mathbf{w}) = \mathbf{H}_1 \mathbf{w}.$$

- (d) Let $\mathbf{V} = (\mathbf{v}_1, \dots, \mathbf{v}_d)$.
 - i. For any $\mathbf{w} \in \mathbb{R}^n$, please find $\mathbf{P}_{\mathbf{V}}(\mathbf{w})$ and the corresponding projection matrix $\mathbf{H}$.
 - ii. Please find $\mathbf{H}$ if we further assume that $\mathbf{v}_i^\top \mathbf{v}_j = 0$, $\forall i \neq j$.
3. (a) Suppose that

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

What are the coordinates of $\mathbf{P}_{\mathbf{A}}(\mathbf{x})$ with respect to the column vectors in $\mathbf{A}$ for any $\mathbf{x} \in \mathbb{R}^2$? Are the coordinates unique?

- (b) Suppose that

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}.$$

What are the coordinates of $\mathbf{P}_{\mathbf{A}}(\mathbf{x})$ with respect to the column vectors in $\mathbf{A}$ for any $\mathbf{x} \in \mathbb{R}^2$? Are the coordinates unique?

4. A matrix $\mathbf{P}$ is called a projection matrix if $\mathbf{P}\mathbf{x}$ is the projection of $\mathbf{x}$ onto $\mathcal{C}(\mathbf{P})$ for any $\mathbf{x}$.

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- (a) Let λ be the eigenvalue of $\mathbf{P}$. Show that λ is either 1 or 0. (*Hint: you may want to figure out what the eigenspaces corresponding to $\lambda = 1$ and $\lambda = 0$ are, respectively.*)
 - (b) Show that $\mathbf{P}$ is a projection matrix if and only if $\mathbf{P}^2 = \mathbf{P}$ and $\mathbf{P}$ is symmetric.
5. Let $\mathbf{B} \in \mathbb{R}^{m \times s}$ and $\mathcal{C}(\mathbf{B})$ be its column space. Suppose that $\mathcal{C}(\mathbf{B})$ is a proper subspace of $\mathcal{C}(\mathbf{A})$. Is $\mathbf{P}_{\mathbf{B}}(\mathbf{x})$ the same as $\mathbf{P}_{\mathbf{B}}(\mathbf{P}_{\mathbf{A}}(\mathbf{x}))$? Please show your claim rigorously.

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Solution 4: Projection

1. ① Strict Convexity:

Define

$$f(\mathbf{z}) = \|\mathbf{x} - \mathbf{z}\|_2^2 = \|\mathbf{z}\|_2^2 + 2\langle \mathbf{x}, \mathbf{z} \rangle + \|\mathbf{x}\|_2^2$$

Then $\nabla^2 f(\mathbf{z}) = 2\mathbf{I}_m > 0 \implies f$ is strictly convex.

The column space $\mathcal{C}(\mathbf{A})$ is convex $\implies \forall \mathbf{a}, \mathbf{b} \in \mathcal{C}(\mathbf{A}), \forall t \in (0, 1), t\mathbf{a} + (1-t)\mathbf{b} \in \mathcal{C}(\mathbf{A})$.

Suppose $\mathbf{z}_1, \mathbf{z}_2 \in \mathcal{C}(\mathbf{A})$ are both minimizers of f over $\mathcal{C}(\mathbf{A})$. Then according to the convexity of f ,

$$f(t\mathbf{z}_1 + (1-t)\mathbf{z}_2) < tf(\mathbf{z}_1) + (1-t)f(\mathbf{z}_2) = f(\mathbf{z}_1) = f(\mathbf{z}_2)$$

contradicting the minimality of $\mathbf{z}_1$ and $\mathbf{z}_2$. Therefore, the minimizer $\mathbf{P}_{\mathbf{A}}(\mathbf{x})$ is unique.

- ② Orthogonality:

Suppose $\mathbf{z}_0$ is a minimizer of f over $\mathcal{C}(\mathbf{A})$. For any $\mathbf{y} \in \mathcal{C}(\mathbf{A})$ and $t \in \mathbb{R}$, define

$$\Phi(t) = \|\mathbf{x} - (\mathbf{z}_0 + t\mathbf{y})\|_2^2 = \|\mathbf{x} - \mathbf{z}_0\|_2^2 - 2t\langle \mathbf{x} - \mathbf{z}_0, \mathbf{y} \rangle + t^2\|\mathbf{y}\|_2^2$$

Notice that $\mathbf{z}_0 + t\mathbf{y} \in \mathcal{C}(\mathbf{A})$.

Since $\mathbf{z}_0$ minimizes f over $\mathcal{C}(\mathbf{A})$, $\Phi(t)$ achieves its minimum at $t = 0 \implies \Phi'(0) = -2\langle \mathbf{x} - \mathbf{z}_0, \mathbf{y} \rangle = 0 \implies \mathbf{x} - \mathbf{z}_0 \perp \mathcal{C}(\mathbf{A})$, i.e., $\mathbf{x} - \mathbf{z}_0 \in \mathcal{C}(\mathbf{A})^\perp$.

Furthermore, if $\mathbf{x} - \mathbf{z}_0 \perp \mathcal{C}(\mathbf{A})$, then for any $\mathbf{y} \in \mathcal{C}(\mathbf{A})$,

$$\|\mathbf{x} - \mathbf{y}\|_2^2 = \|\mathbf{x} - \mathbf{z}_0 + \mathbf{z}_0 - \mathbf{y}\|_2^2 = \|\mathbf{x} - \mathbf{z}_0\|_2^2 + \|\mathbf{z}_0 - \mathbf{y}\|_2^2 \geq \|\mathbf{x} - \mathbf{z}_0\|_2^2$$

$\implies \mathbf{z}_0$ is a minimizer of f over $\mathcal{C}(\mathbf{A})$.

If $\mathbf{z}_1, \mathbf{z}_2$ both satisfy $\mathbf{x} - \mathbf{z}_i \perp \mathcal{C}(\mathbf{A})$ ($i = 1, 2$), then $\mathbf{z}_1 - \mathbf{z}_2 \in \mathcal{C}(\mathbf{A})$ and $\mathbf{z}_1 - \mathbf{z}_2 = (\mathbf{x} - \mathbf{z}_2) - (\mathbf{x} - \mathbf{z}_1) \perp \mathcal{C}(\mathbf{A}) \implies \mathbf{z}_1 - \mathbf{z}_2 = \mathbf{0} \implies \mathbf{z}_1 = \mathbf{z}_2$, i.e. the minimizer $\mathbf{P}_{\mathbf{A}}(\mathbf{x})$ is unique.

2. (a)

$$\mathbf{P}_{\mathbf{v}_1}(\mathbf{w}) = \underset{\substack{\alpha \mathbf{v}_1 \\ \alpha \in \mathbb{R}, \mathbf{v}_1 \in \mathbb{R}^n}}{\operatorname{argmin}} \|\mathbf{w} - \alpha \mathbf{v}_1\|_2$$

$$\frac{d}{d\alpha} \|\mathbf{w} - \alpha \mathbf{v}_1\|_2^2 = \frac{d}{d\alpha} (\mathbf{w} - \alpha \mathbf{v}_1)^\top (\mathbf{w} - \alpha \mathbf{v}_1) = -2\mathbf{v}_1^\top (\mathbf{w} - \alpha \mathbf{v}_1) = 0$$

$$\implies \alpha^* = \frac{\mathbf{v}_1^\top \mathbf{w}}{\mathbf{v}_1^\top \mathbf{v}_1} \implies \mathbf{P}_{\mathbf{v}_1}(\mathbf{w}) = \frac{\mathbf{v}_1^\top \mathbf{w}}{\mathbf{v}_1^\top \mathbf{v}_1} \mathbf{v}_1$$

- (b) According to the result in (a),

$$\begin{aligned} \mathbf{P}_{\mathbf{v}_1}(\alpha \mathbf{u} + \beta \mathbf{w}) &= \frac{\mathbf{v}_1^\top (\alpha \mathbf{u} + \beta \mathbf{w})}{\mathbf{v}_1^\top \mathbf{v}_1} \mathbf{v}_1 = \alpha \frac{\mathbf{v}_1^\top \mathbf{u}}{\mathbf{v}_1^\top \mathbf{v}_1} \mathbf{v}_1 + \beta \frac{\mathbf{v}_1^\top \mathbf{w}}{\mathbf{v}_1^\top \mathbf{v}_1} \mathbf{v}_1 \\ &= \alpha \mathbf{P}_{\mathbf{v}_1}(\mathbf{u}) + \beta \mathbf{P}_{\mathbf{v}_1}(\mathbf{w}) \end{aligned}$$

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(c)

$$\mathbf{P}_{\mathbf{v}_1}(\mathbf{w}) = \frac{\mathbf{v}_1^T \mathbf{w}}{\mathbf{v}_1^T \mathbf{v}_1} \mathbf{v}_1 = \left(\frac{\mathbf{v}_1 \mathbf{v}_1^T}{\mathbf{v}_1^T \mathbf{v}_1} \right) \mathbf{w} := \mathbf{H}_1 \mathbf{w}$$

(d) i.

$$\mathbf{P}_{\mathbf{V}}(\mathbf{w}) = \underset{\substack{\mathbf{y} \in \mathbb{R}^d, \mathbf{V} \in \mathbb{R}^{n \times d}}}{\operatorname{argmin}} \|\mathbf{w} - \mathbf{V}\mathbf{y}\|_2$$

$$\frac{d}{d\mathbf{y}} \|\mathbf{w} - \mathbf{V}\mathbf{y}\|_2^2 = \frac{d}{d\mathbf{y}} (\mathbf{w} - \mathbf{V}\mathbf{y})^T (\mathbf{w} - \mathbf{V}\mathbf{y}) = -2\mathbf{V}^T (\mathbf{w} - \mathbf{V}\mathbf{y}) = 0$$

$\mathbf{v}_i, i = 1, \dots, d$ are linearly independent $\implies \mathbf{V}$ has full column rank $\implies \mathbf{V}^T \mathbf{V}$ is invertible.

$$\implies \mathbf{y}^* = (\mathbf{V}^T \mathbf{V})^{-1} \mathbf{V}^T \mathbf{w} \implies \mathbf{P}_{\mathbf{V}}(\mathbf{w}) = \mathbf{V}(\mathbf{V}^T \mathbf{V})^{-1} \mathbf{V}^T \mathbf{w}$$

$$\implies \mathbf{H} = \mathbf{V}(\mathbf{V}^T \mathbf{V})^{-1} \mathbf{V}^T$$

ii.

$$\begin{aligned} \mathbf{v}_i^T \mathbf{v}_j = 0, \forall i \neq j \implies \mathbf{V}^T \mathbf{V} &= \begin{bmatrix} \mathbf{v}_1^T \mathbf{v}_1 & 0 & \cdots & 0 \\ 0 & \mathbf{v}_2^T \mathbf{v}_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathbf{v}_d^T \mathbf{v}_d \end{bmatrix} \\ \implies (\mathbf{V}^T \mathbf{V})^{-1} &= \begin{bmatrix} \frac{1}{\mathbf{v}_1^T \mathbf{v}_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\mathbf{v}_2^T \mathbf{v}_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\mathbf{v}_d^T \mathbf{v}_d} \end{bmatrix} \\ \implies \mathbf{H} &= \sum_{i=1}^d \frac{\mathbf{v}_i \mathbf{v}_i^T}{\mathbf{v}_i^T \mathbf{v}_i} \end{aligned}$$

3. (a)

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \implies \mathbf{x} \in \mathcal{C}(\mathbf{A}) = \mathbb{R}^2 \implies \mathbf{P}_{\mathbf{A}}(\mathbf{x}) = \mathbf{x} = \mathbf{A} \cdot \mathbf{x}$$

The coordinates of $\mathbf{P}_{\mathbf{A}}(\mathbf{x})$ with respect to the column vectors in $\mathbf{A}$ are unique and equal to $\mathbf{x}$ itself.

(b)

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \implies \mathcal{C}(\mathbf{A}) = \{\alpha(1, 1)^T : \alpha \in \mathbb{R}\}$$

$$\mathbf{P}_{\mathbf{A}}(\mathbf{x}) = \underset{\substack{\alpha(1,1)^T \\ \alpha \in \mathbb{R}}}{\operatorname{argmin}} \|\mathbf{x} - \alpha(1, 1)^T\|_2$$

$$\xrightarrow{2.(a)} \mathbf{P}_{\mathbf{A}}(\mathbf{x}) = \frac{(1, 1)\mathbf{x}}{(1, 1)(1, 1)^T} (1, 1)^T = \frac{x_1 + x_2}{2} (1, 1)^T = \mathbf{A} \cdot (c_1, c_2)^T$$

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$\implies$ The set of all coordinates of $\mathbf{P}_A(\mathbf{x})$ with respect to the column vectors in $\mathbf{A}$ is the affine line:

$$\left\{ (c_1, c_2)^T \in \mathbb{R}^2 : c_1 + 2c_2 = \frac{x_1 + x_2}{2} \right\}$$

Thus the coordinates are not unique.

4. (a) Let λ be an eigenvalue of $\mathbf{P}$ and $\mathbf{v}$ be the corresponding eigenvector. Then $\mathbf{P}\mathbf{v} = \lambda\mathbf{v}$. Since $\mathbf{P}$ is a projection matrix, $\mathbf{P}\mathbf{v}$ is the projection of $\mathbf{v}$ onto $\mathcal{C}(\mathbf{P})$
 $\implies \mathbf{v} = \mathbf{P}\mathbf{v} + (\mathbf{v} - \mathbf{P}\mathbf{v})$, where $\mathbf{P}\mathbf{v} \in \mathcal{C}(\mathbf{P})$ and $\mathbf{v} - \mathbf{P}\mathbf{v} \in \mathcal{C}(\mathbf{P})^\perp$.

$$\implies \|\mathbf{v}\|_2^2 = \|\mathbf{P}\mathbf{v}\|_2^2 + \|\mathbf{v} - \mathbf{P}\mathbf{v}\|_2^2 \xrightarrow{\mathbf{P}\mathbf{v}=\lambda\mathbf{v}} (\lambda^2 - \lambda) \|\mathbf{v}\|_2^2 = 0 \implies \lambda \in \{0, 1\}$$

- (b) ① Necessity:

$\forall \mathbf{x} \in \mathbb{R}^m, \mathbf{P}\mathbf{x} \in \mathcal{C}(\mathbf{P})$. We need to prove that $\mathbf{x} - \mathbf{P}\mathbf{x} \in \mathcal{C}(\mathbf{P})^\perp$.

$\forall \mathbf{y} \in \mathcal{C}(\mathbf{P}), \exists \mathbf{z} \in \mathbb{R}^m$, s.t. $\mathbf{y} = \mathbf{P}\mathbf{z}$. Then

$$\begin{aligned} \langle \mathbf{x} - \mathbf{P}\mathbf{x}, \mathbf{y} \rangle &= \langle \mathbf{x} - \mathbf{P}\mathbf{x}, \mathbf{P}\mathbf{z} \rangle = \langle \mathbf{P}^T(\mathbf{x} - \mathbf{P}\mathbf{x}), \mathbf{z} \rangle \\ &\stackrel{\mathbf{P}^T=\mathbf{P}}{=} \langle \mathbf{P}(\mathbf{x} - \mathbf{P}\mathbf{x}), \mathbf{z} \rangle \\ &\stackrel{\mathbf{P}^2=\mathbf{P}}{=} \langle \mathbf{P}\mathbf{x} - \mathbf{P}^2\mathbf{x}, \mathbf{z} \rangle = 0 \end{aligned}$$

$\implies \mathbf{x} - \mathbf{P}\mathbf{x} \in \mathcal{C}(\mathbf{P})^\perp \implies \mathbf{P}$ is a projection matrix.

② Sufficiency:

$\mathbf{P}$ is a projection matrix $\implies \forall \mathbf{x} \in \mathbb{R}^m, \mathbf{P}\mathbf{x} \in \mathcal{C}(\mathbf{P})$

$\implies \mathbf{P}(\mathbf{P}\mathbf{x}) = \mathbf{argmin}_{\mathbf{z} \in \mathcal{C}(\mathbf{P})} \|\mathbf{P}\mathbf{x} - \mathbf{z}\|_2 = \mathbf{P}\mathbf{x} \implies \mathbf{P}^2 = \mathbf{P}$.

$\mathbf{P}$ is a projection matrix $\implies \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^m, \mathbf{x} = \mathbf{P}\mathbf{x} + (\mathbf{x} - \mathbf{P}\mathbf{x})$, where $\mathbf{P}\mathbf{x} \in \mathcal{C}(\mathbf{P})$ and $\mathbf{x} - \mathbf{P}\mathbf{x} \in \mathcal{C}(\mathbf{P})^\perp$. Since $\mathbf{P}\mathbf{y} \in \mathcal{C}(\mathbf{P})$, we have

$$\langle \mathbf{P}\mathbf{x}, \mathbf{y} - \mathbf{P}\mathbf{y} \rangle = 0 \implies \langle \mathbf{P}\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{P}\mathbf{x}, \mathbf{P}\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{P}\mathbf{y} \rangle = \langle \mathbf{P}^T\mathbf{x}, \mathbf{y} \rangle$$

$$\implies \mathbf{P}^T = \mathbf{P}.$$

5. $\forall \mathbf{x} \in \mathbb{R}^m$,

$\mathcal{C}(\mathbf{B})$ is a proper subspace of $\mathcal{C}(\mathbf{A}) \implies \forall \mathbf{z} \in \mathcal{C}(\mathbf{P}), \mathbf{z} \in \mathcal{C}(\mathbf{A}) \implies \mathbf{P}_A(\mathbf{x}) - \mathbf{z} \in \mathcal{C}(\mathbf{A})$

$$\begin{aligned} \mathbf{x} - \mathbf{P}_A(\mathbf{x}) \in \mathcal{C}(\mathbf{A})^\perp &\implies \|\mathbf{x} - \mathbf{z}\|_2^2 = \|(\mathbf{P}_A(\mathbf{x}) - \mathbf{z}) + (\mathbf{x} - \mathbf{P}_A(\mathbf{x}))\|_2^2 \\ &= \|\mathbf{P}_A(\mathbf{x}) - \mathbf{z}\|_2^2 + \|\mathbf{x} - \mathbf{P}_A(\mathbf{x})\|_2^2 \end{aligned}$$

$$\implies \mathbf{argmin}_{\mathbf{z} \in \mathcal{C}(\mathbf{B})} \|\mathbf{x} - \mathbf{z}\|_2 = \mathbf{argmin}_{\mathbf{z} \in \mathcal{C}(\mathbf{B})} \|\mathbf{P}_A(\mathbf{x}) - \mathbf{z}\|_2 \implies \mathbf{P}_B(\mathbf{x}) = \mathbf{P}_B(\mathbf{P}_A(\mathbf{x}))$$

■

Homework 1

Exercise 5: Derivatives with matrices

Definition 1 (Differentiability). [?] Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a function, $\mathbf{x}_0 \in \mathbb{R}^n$ be a point, and let $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. We say that f is *differentiable at $\mathbf{x}_0$ with derivative L* if we have

$$\lim_{\mathbf{x} \rightarrow \mathbf{x}_0; \mathbf{x} \neq \mathbf{x}_0} \frac{\|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)\|_2}{\|\mathbf{x} - \mathbf{x}_0\|_2} = 0.$$

We denote this derivative by $f'(\mathbf{x}_0)$.

1. Let $\mathbf{x}, \mathbf{a} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^m$. Consider the functions as follows. Please show that they are differentiable and find $f'(\mathbf{x})$.
 - (a) $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$.
 - (b) $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{x}$.
2. Consider a differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. The **Jacobian Matrix with denominator layout** is defined by:

$$\frac{\partial f}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial f_1(\mathbf{x})}{\partial x_1} & \frac{\partial f_2(\mathbf{x})}{\partial x_1} & \cdots & \frac{\partial f_m(\mathbf{x})}{\partial x_1} \\ \frac{\partial f_1(\mathbf{x})}{\partial x_2} & \frac{\partial f_2(\mathbf{x})}{\partial x_2} & \cdots & \frac{\partial f_m(\mathbf{x})}{\partial x_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_1(\mathbf{x})}{\partial x_n} & \frac{\partial f_2(\mathbf{x})}{\partial x_n} & \cdots & \frac{\partial f_m(\mathbf{x})}{\partial x_n} \end{bmatrix}.$$

Please show that

$$L(\mathbf{x} - \mathbf{x}_0) = \left(\frac{\partial f}{\partial \mathbf{x}} \right)^\top (\mathbf{x} - \mathbf{x}_0),$$

where $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the derivative in Definition 1.

3. Please follow Definition 1 and give the definition of the differentiability of the functions $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$.
4. Let $f(\mathbf{X}) = \text{tr}(\mathbf{A}^\top \mathbf{X})$, where $\mathbf{A}, \mathbf{X} \in \mathbb{R}^{n \times m}$, and $\text{tr}(\cdot)$ denotes the trace of a matrix. Please discuss the differentiability of f and find f' if it is differentiable.
5. (Optional) Let $f(\mathbf{X}) = \det(\mathbf{X})$, where $\det(\mathbf{X})$ is the determinant of $\mathbf{X} \in \mathbb{R}^{n \times n}$. Please discuss the differentiability of f rigorously according to your definition in the last part. If f is differentiable, please find $f'(\mathbf{X})$.
6. (Optional) Let $\mathbf{S}_{++}^n$ be the space of all positive definite $n \times n$ matrices. Please show the function $f : \mathbf{S}_{++}^n \rightarrow \mathbb{R}$ defined by $f(\mathbf{X}) = \text{tr} \mathbf{X}^{-1}$ is differentiable on $\mathbf{S}_{++}^n$. (Hint: Expand the expression $(\mathbf{X} + t\mathbf{Y})^{-1}$ as a power series.)

Homework 1

Solution 5: Derivatives with matrices

1. (a) Let $L(\mathbf{h}) = \mathbf{a}^\top \mathbf{h}$, $\forall \mathbf{h} \in \mathbb{R}^n$, then L is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}$.
For any $\mathbf{x}_0 \in \mathbb{R}^n$ and $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x} \neq \mathbf{x}_0$,

$$\begin{aligned} \frac{|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)|}{\|\mathbf{x} - \mathbf{x}_0\|_2} &= \frac{|\mathbf{a}^\top \mathbf{x} - \mathbf{a}^\top \mathbf{x}_0 - \mathbf{a}^\top (\mathbf{x} - \mathbf{x}_0)|}{\|\mathbf{x} - \mathbf{x}_0\|_2} \\ &= \frac{0}{\|\mathbf{x} - \mathbf{x}_0\|_2} = 0 \end{aligned}$$

$\implies f$ is differentiable at $\mathbf{x}_0$ with derivative L and $f'(\mathbf{x}) = \mathbf{a}^\top$.

- (b) Let $L(\mathbf{h}) = 2\mathbf{x}_0^\top \mathbf{h}$, $\forall \mathbf{h} \in \mathbb{R}^n$, then L is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}$.
For any $\mathbf{x}_0 \in \mathbb{R}^n$ and $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x} \neq \mathbf{x}_0$,

$$\begin{aligned} \frac{|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)|}{\|\mathbf{x} - \mathbf{x}_0\|_2} &= \frac{|\mathbf{x}^\top \mathbf{x} - \mathbf{x}_0^\top \mathbf{x}_0 - 2\mathbf{x}_0^\top (\mathbf{x} - \mathbf{x}_0)|}{\|\mathbf{x} - \mathbf{x}_0\|_2} \\ &= \frac{|(\mathbf{x} - \mathbf{x}_0)^\top (\mathbf{x} - \mathbf{x}_0)|}{\|\mathbf{x} - \mathbf{x}_0\|_2} \\ &= \frac{\|\mathbf{x} - \mathbf{x}_0\|_2^2}{\|\mathbf{x} - \mathbf{x}_0\|_2} \\ &= \|\mathbf{x} - \mathbf{x}_0\|_2 \end{aligned}$$

$\implies \lim_{\mathbf{x} \rightarrow \mathbf{x}_0; \mathbf{x} \neq \mathbf{x}_0} \frac{|f(\mathbf{x}) - f(\mathbf{x}_0) - L(\mathbf{x} - \mathbf{x}_0)|}{\|\mathbf{x} - \mathbf{x}_0\|_2} = 0 \implies f$ is differentiable at $\mathbf{x}_0$ with derivative L and $f'(\mathbf{x}) = 2\mathbf{x}^\top$.

Key point: Observe the linear term of $\mathbf{x} - \mathbf{x}_0$ in $f(\mathbf{x}) - f(\mathbf{x}_0)$.

2. Denote $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ as the standard basis of $\mathbb{R}^n$.

For a fixed $i \in \{1, \dots, n\}$, we have

$$\lim_{t \rightarrow 0} \frac{f(\mathbf{x}_0 + t\mathbf{e}_i) - f(\mathbf{x}_0)}{t} = \left(\frac{\partial f_1}{\partial x_i}(\mathbf{x}_0), \frac{\partial f_2}{\partial x_i}(\mathbf{x}_0), \dots, \frac{\partial f_m}{\partial x_i}(\mathbf{x}_0) \right)^\top = L\mathbf{e}_i$$

which is the i -th column of

$$\frac{\partial f}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial f_1(\mathbf{x})}{\partial x_1} & \frac{\partial f_2(\mathbf{x})}{\partial x_1} & \dots & \frac{\partial f_m(\mathbf{x})}{\partial x_1} \\ \frac{\partial f_1(\mathbf{x})}{\partial x_2} & \frac{\partial f_2(\mathbf{x})}{\partial x_2} & \dots & \frac{\partial f_m(\mathbf{x})}{\partial x_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_1(\mathbf{x})}{\partial x_n} & \frac{\partial f_2(\mathbf{x})}{\partial x_n} & \dots & \frac{\partial f_m(\mathbf{x})}{\partial x_n} \end{bmatrix}.$$

Homework 1

For any $\mathbf{h} = \sum_{i=1}^n h_i \mathbf{e}_i$, we have

$$L\mathbf{h} = L\left(\sum_{i=1}^n h_i \mathbf{e}_i\right) = \sum_{i=1}^n h_i L\mathbf{e}_i = \sum_{i=1}^n h_i \left[\left(\frac{\partial f}{\partial \mathbf{x}}(\mathbf{x}_0)\right)^\top\right]_{\cdot, i} = \left(\frac{\partial f}{\partial \mathbf{x}}(\mathbf{x}_0)\right)^\top \mathbf{h}$$

Let $\mathbf{h} = \mathbf{x} - \mathbf{x}_0$, we conclude

$$L(\mathbf{x} - \mathbf{x}_0) = \left(\frac{\partial f}{\partial \mathbf{x}}\right)^\top (\mathbf{x} - \mathbf{x}_0).$$

3. Let $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ be a function, $\mathbf{X}_0 \in \mathbb{R}^{n \times n}$ be a point, and let $L : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ be a linear transformation. We say that f is *differentiable at $\mathbf{X}_0$ with derivative L* if we have

$$\lim_{\mathbf{X} \rightarrow \mathbf{X}_0; \mathbf{X} \neq \mathbf{X}_0} \frac{|f(\mathbf{X}) - f(\mathbf{X}_0) - L(\mathbf{X} - \mathbf{X}_0)|}{\|\mathbf{X} - \mathbf{X}_0\|_2} = 0.$$

We denote this derivative by $f'(\mathbf{X}_0)$.

4. $\forall \alpha, \beta > 0$ and $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{n \times m}$,

$$f(\alpha \mathbf{X} + \beta \mathbf{Y}) = \text{tr}\left(\mathbf{A}^\top (\alpha \mathbf{X} + \beta \mathbf{Y})\right) = \alpha \text{tr}\left(\mathbf{A}^\top \mathbf{X}\right) + \beta \text{tr}\left(\mathbf{A}^\top \mathbf{Y}\right) = \alpha f(\mathbf{X}) + \beta f(\mathbf{Y})$$

$\implies f$ is linear $\implies$ let $L(\mathbf{H}) = \text{tr}(\mathbf{A}^\top \mathbf{H})$. It is clear that $f(\mathbf{X}) - f(\mathbf{X}_0) - L(\mathbf{X} - \mathbf{X}_0) = 0$
 $\implies f$ is differentiable and $f'(\mathbf{X}) = \text{tr}(\mathbf{A}^\top \mathbf{X})$.

5.

6.

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Homework 1

Exercise 6: Linear Space

1. Let $P_n[x]$ be the set of all polynomials on $\mathbb{R}$ with degree at most n . Show that $P_n[x]$ is a linear space.
2. A real symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is called *positive definite*, written $\mathbf{A} \succ \mathbf{0}$, if for all $\mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq \mathbf{0}$,
$$\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0.$$

Let the set of all positive definite matrices be

$$\mathbb{S}_{++}^n := \left\{ \mathbf{A} \in \mathbb{R}^{n \times n} : \mathbf{A} = \mathbf{A}^\top, \mathbf{x}^\top \mathbf{A} \mathbf{x} > 0 \text{ for all } \mathbf{x} \neq \mathbf{0} \right\}.$$

Is $\mathbb{S}_{++}^n$ a linear subspace of $\mathbb{R}^{n \times n}$? Please show your conclusion in detail.

Homework 1

Solution 6: Linear Space

1. $\forall p(x) = \sum_{k=0}^m a_k x^k, q(x) = \sum_{k=0}^m b_k x^k, r(x) = \sum_{k=0}^m c_k x^k, a_k, b_k, c_k \in \mathbb{R}, 0 \leq k \leq n, \forall \alpha, \beta \in \mathbb{R}$:

(a) Closure under addition:

$$p(x) + q(x) = \sum_{k=0}^m (a_k + b_k) x^k \in P_n[x]$$

(b) Associativity of addition:

$$(p + q) + r = \sum_{k=0}^m [(a_k + b_k) + c_k] x^k = \sum_{k=0}^m [a_k + (b_k + c_k)] x^k = p + (q + r)$$

(c) Commutativity of addition:

$$p(x) + q(x) = \sum_{k=0}^m (a_k + b_k) x^k = \sum_{k=0}^m (b_k + a_k) x^k = q(x) + p(x)$$

(d) Additive identity:

$$0 \in P_n[x] \text{ and } p + 0 = 0 + p$$

(e) Additive Inverse:

$$-p = \sum_{k=0}^m (-a_k) x^k \in P_n[x] \text{ and } p + (-p) = 0$$

(f) Closure under scalar multiplication:

$$\alpha p = \sum_{k=0}^m (\alpha a_k) x^k \in P_n[x]$$

(g) Distributivity of scalar over vector addition:

$$\alpha(p + q) = \sum_{k=0}^m [\alpha(a_k + b_k)] x^k = \sum_{k=0}^m (\alpha a_k + \alpha b_k) x^k = \alpha p + \alpha q$$

(h) Distributivity of scalar addition over vector:

$$(\alpha + \beta)p = \sum_{k=0}^m [(\alpha + \beta)a_k] x^k = \sum_{k=0}^m (\alpha a_k + \beta a_k) x^k = \alpha p + \beta p$$

Homework 1

(i) Compatibility of scalar multiplication:

$$\alpha(\beta p) = \sum_{k=0}^m [\alpha(\beta a_k)] x^k = \sum_{k=0}^m [(\alpha\beta) a_k] x^k = (\alpha\beta) p$$

(j) Unit scalar:

$$1 \in P_n[x] \text{ and } 1 \cdot p = p \cdot 1 = p$$

2. No.

(a)

$$\forall \mathbf{x} \in \mathbb{R}^n, \mathbf{x}^\top \mathbf{0} \mathbf{x} = 0 \implies \mathbf{0} \notin \mathbb{S}_{++}^n$$

(b)

$$\forall \alpha < 0, \mathbf{A} \in \mathbb{S}_{++}^n, \mathbf{x}^\top (\alpha \mathbf{A}) \mathbf{x} < 0 \implies \alpha \mathbf{A} \notin \mathbb{S}_{++}^n$$

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Homework 1

Exercise 7: Basis and Coordinates

Suppose that $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ is a basis of an n -dimensional vector space V .

1. Show that $\{\lambda_1 \mathbf{a}_1, \lambda_2 \mathbf{a}_2, \dots, \lambda_n \mathbf{a}_n\}$ is also a basis of V for nonzero scalars $\lambda_1, \lambda_2, \dots, \lambda_n$.
2. Let $V = \mathbb{R}^n$, $\mathbf{A} = (\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n) \in \mathbb{R}^{n \times n}$ and $\mathbf{B} = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n) \in \mathbb{R}^{n \times n}$. $(\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n) = (\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n)\mathbf{P}$, where $\mathbf{P} \in \mathbb{R}^{n \times n}$ and $\mathbf{b}_i \in \mathbb{R}^n$, for any $i \in \{1, \dots, n\}$. Show that $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ is also a basis of V for any invertible matrix $\mathbf{P}$.
3. Suppose that the coordinate of a vector $\mathbf{v}$ under the basis $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ is $\mathbf{x} = (x_1, x_2, \dots, x_n)$.
 - (a) What is the coordinate of $\mathbf{v}$ under $\{\lambda_1 \mathbf{a}_1, \lambda_2 \mathbf{a}_2, \dots, \lambda_n \mathbf{a}_n\}$?
 - (b) What are the coordinates of $\mathbf{w} = \mathbf{a}_1 + \dots + \mathbf{a}_n$ under $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ and $\{\lambda_1 \mathbf{a}_1, \lambda_2 \mathbf{a}_2, \dots, \lambda_n \mathbf{a}_n\}$? Note that $\lambda_i \neq 0$ for any $i \in \{1, \dots, n\}$.
4. Suppose $\mathbf{a} = (1, 0)$, $\mathbf{b} = (0, 1)$ and $\mathbf{c} = (-1, 0)$ are three unit vectors in two-dimensional space. $\mathbf{v} = (x, y)$ is a vector in two-dimensional space.
 - (a) Please find the coordinate of $\mathbf{v}$ under basis $\{\mathbf{c}, \mathbf{b}\}$? Is the coordinate unique?
 - (b) Please find all the possible combination coefficients of $\mathbf{v}$ under vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, i.e., $\mathbf{v} = x'\mathbf{a} + y'\mathbf{b} + z'\mathbf{c}$.
 - (c) (**Bonus**) Each set of combination coefficients (x', y', z') in (b) forms a vector in $\mathbb{R}^3$. Please find the combination coefficients with minimum ℓ_1 -norm.

Homework 1

Solution 7: Basis and Coordinates

1. Suppose $\sum_{i=1}^n c_i(\lambda_i \mathbf{a}_i) = \sum_{i=1}^n (c_i \lambda_i) \mathbf{a}_i = 0$. Because $\{\mathbf{a}_i\}_{i=1}^n$ is linear independent, we must have $c_i \lambda_i = 0 \xrightarrow{\lambda_i \neq 0} c_i = 0$ for all i , i.e., $\{\lambda_i \mathbf{a}_i\}$ is linear independent.

$\forall \mathbf{v} \in \mathbf{V}$, $\{\mathbf{a}_i\}$ is a basis $\implies$ there exists unique scalars $\{\alpha_i\}$ such that

$$\mathbf{v} = \sum_{i=1}^n \alpha_i \mathbf{a}_i = \sum_{i=1}^n \left(\frac{\alpha_i}{\lambda_i} \right) (\lambda_i \alpha_i).$$

Thus $\mathbf{v}$ is a linear combination for the vectors in $\{\lambda_i \mathbf{a}_i\}$.

In conclusion, $\{\lambda_i \mathbf{a}_i\}$ is also a basis of an n -dimensional vector space V .

2. ① $|\mathbf{B}| = |\mathbf{A}\mathbf{P}| = |\mathbf{A}| \cdot |\mathbf{P}| \neq 0 \implies \mathbf{B}$ is invertible $\implies$ the columns $\mathbf{b}_i$, $1 \leq i \leq n$ are linearly independent. Being n independent vectors in $\mathbb{R}^n$, they span $\mathbb{R}^n$, i.e., $\{\mathbf{b}_i\}$, $1 \leq i \leq n$ is also a basis of V for any invertible matrix $\mathbf{P}$.

② $\forall \mathbf{x} \in \mathbb{R}^n$, there exists a unique $\mathbf{c} \in \mathbb{R}^n$ with $\mathbf{x} = \mathbf{A}\mathbf{c} \xrightarrow{\mathbf{P} \text{ is invertible}} \mathbf{x} = \mathbf{A}\mathbf{c} = \mathbf{A}\mathbf{P}\mathbf{P}^{-1}\mathbf{c} = \mathbf{B}(\mathbf{P}^{-1}\mathbf{c}) \implies \{\mathbf{b}_i\}$, $1 \leq i \leq n$ is also a basis of V for any invertible matrix $\mathbf{P}$.

3. (a) Set $\mathbf{y} = (y_1, y_2, \dots, y_n)$ as the coordinate of $\mathbf{v}$ under $\{\lambda_i \mathbf{a}_i\}$, $1 \leq i \leq n$.

$$\mathbf{v} = \sum_{i=1}^n y_i (\lambda_i \mathbf{a}_i) = \sum_{i=1}^n x_i \mathbf{a}_i \implies y_i = \frac{x_i}{\lambda_i}$$

Therefore, the coordinate of $\mathbf{v}$ under $\{\lambda_i \mathbf{a}_i\}$, $1 \leq i \leq n$ is $\mathbf{y} = \left(\frac{x_1}{\lambda_1}, \frac{x_2}{\lambda_2}, \dots, \frac{x_n}{\lambda_n} \right)$.

(b) It is clear that $\mathbf{x} = (1, 1, \dots, 1)^\top$ and $\mathbf{y} = \left(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n} \right)$.

4. (a) It is clear that the coordinate of $\mathbf{v}$ under basis $\{\mathbf{c}, \mathbf{b}\}$ is $(-x, y)$. And the coordinate is unique because $\{\mathbf{c}, \mathbf{b}\}$ is a basis of $\mathbb{R}^2$.

(b) $\mathbf{v} = x'\mathbf{a} + y'\mathbf{b} + z'\mathbf{c} = (x' - z', y') = (x, y) \implies (x', y', z') = (x + t, y, t), \forall t \in \mathbb{R}$.

(c)

$$\min \|(x', y', z')\|_1 = \min |x'| + |y'| + |z'| = \min |x'| + |z'| = \min |x + t| + |t|$$

$$\begin{cases} |x + t| + |t| \geq |x + t - t| = |x| \\ |x + t| + |t| = |x| \text{ when } t \in [-x, 0] \text{ and } x \geq 0 \text{ or } t \in [0, -x] \text{ and } x < 0 \end{cases}$$

In conclusion, $\min \|(x', y', z')\|_1 = |x| + |y|$.

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Homework 1

Exercise 8: Rank of matrices

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$.

1. Please show that

(a) $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\top) = \text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{A} \mathbf{A}^\top)$;

(b) $\text{rank}(\mathbf{AB}) \leq \text{rank}(\mathbf{A})$; (please give an example when the equality holds)

2. The *column space* of $\mathbf{A}$ is defined by

$$\mathcal{C}(\mathbf{A}) = \{\mathbf{y} \in \mathbb{R}^m : \mathbf{y} = \mathbf{Ax}, \mathbf{x} \in \mathbb{R}^n\}.$$

The *null space* of $\mathbf{A}$ is defined by

$$\mathcal{N}(\mathbf{A}) = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} = \mathbf{0}\}.$$

Notice that, the rank of $\mathbf{A}$ is the dimension of the column space of $\mathbf{A}$.

Please show that

(a) $\text{rank}(\mathbf{A}) = \dim(\mathcal{C}(\mathbf{A}))$;

(b) $\text{rank}(\mathbf{A}) + \dim(\mathcal{N}(\mathbf{A})) = n$.

3. Given that

$$\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{B}) - \dim(\mathcal{C}(\mathbf{B}) \cap \mathcal{N}(\mathbf{A})). \quad (1)$$

Please show the results in 1.(b) by Eq. (1).

Homework 1

Solution 8: Rank of matrices

1. (a) The column rank of a matrix equals its row rank $\implies \mathbf{rank}(\mathbf{A}) = \mathbf{rank}(\mathbf{A})$.
For $\mathbf{x} \in \mathbb{R}^n$,

$$\begin{cases} (\mathbf{Ax} = \mathbf{0} \implies \mathbf{A}^T \mathbf{Ax} = \mathbf{0}) \implies \mathcal{N}(\mathbf{A}^T \mathbf{A}) \supset \mathcal{N}(\mathbf{A}) \\ (\mathbf{A}^T \mathbf{Ax} = \mathbf{0} \implies \|\mathbf{Ax}\|_2^2 = \mathbf{x}^T \mathbf{A}^T \mathbf{Ax} = \mathbf{0} \implies \mathbf{Ax} = \mathbf{0}) \implies \mathcal{N}(\mathbf{A}^T \mathbf{A}) \subset \mathcal{N}(\mathbf{A}) \end{cases}$$

$$\implies \mathcal{N}(\mathbf{A}^T \mathbf{A}) = \mathcal{N}(\mathbf{A})$$

$$\implies \mathbf{rank}(\mathbf{A}^T \mathbf{A}) = n - \dim \mathcal{N}(\mathbf{A}^T \mathbf{A}) = n - \dim \mathcal{N}(\mathbf{A}) = \mathbf{rank}(\mathbf{A})$$

In the same way, $\mathbf{rank}(\mathbf{AA}^T) = \mathbf{rank}(\mathbf{A})$.

In conclusion, $\mathbf{rank}(\mathbf{A}) = \mathbf{rank}(\mathbf{A}^T) = \mathbf{rank}(\mathbf{A}^T \mathbf{A}) = \mathbf{rank}(\mathbf{AA}^T)$.

- (b) $\text{Im}(\mathbf{AB}) = \mathbf{A}(\text{Im}(\mathbf{B})) \subset \text{Im}(\mathbf{A}) \implies \mathbf{rank}(\mathbf{AB}) = \dim(\text{Im}(\mathbf{AB})) \leq \dim(\text{Im}(\mathbf{A})) = \mathbf{rank}(\mathbf{A})$.

E.g.:

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ and $\mathbf{B} = I_n$. Then $\mathbf{AB} = \mathbf{A} \implies \mathbf{rank}(\mathbf{AB}) = \mathbf{rank}(\mathbf{A})$.

2. (a) By definition, $\mathbf{rank}(\mathbf{A})$ is the maximal number of linearly independent columns of $\mathbf{A}$ and the dimension of the span of a finite set of vectors equals the maximal size of a linearly independent subset of that set. Therefore, $\mathbf{rank}(\mathbf{A}) = \dim(\mathcal{C}(\mathbf{A}))$.
- (b) Let $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ be a basis of $\mathcal{N}(\mathbf{A})$, where $k = \dim(\mathcal{N}(\mathbf{A}))$. We can extend this basis to a basis of $\mathbb{R}^n$, i.e., there exist vectors $\{\mathbf{w}_1, \dots, \mathbf{w}_{n-k}\}$ such that $\{\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}_1, \dots, \mathbf{w}_{n-k}\}$ is a basis of $\mathbb{R}^n$.

We claim that $\{\mathbf{Aw}_1, \dots, \mathbf{Aw}_{n-k}\}$ is a basis of $\mathcal{C}(\mathbf{A})$.

Spanning: $\forall \mathbf{y} \in \mathcal{C}(\mathbf{A})$, there exists $\mathbf{x} \in \mathbb{R}^n$ such that $\mathbf{y} = \mathbf{Ax}$.

Since $\{\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}_1, \dots, \mathbf{w}_{n-k}\}$ is a basis of $\mathbb{R}^n$, there exist scalars $\{\alpha_i\}_{i=1}^k$ and $\{\beta_j\}_{j=1}^{n-k}$ such that

$$\mathbf{x} = \sum_{i=1}^k \alpha_i \mathbf{v}_i + \sum_{j=1}^{n-k} \beta_j \mathbf{w}_j$$

$$\implies \mathbf{y} = \mathbf{Ax} = \sum_{i=1}^k \alpha_i \mathbf{Av}_i + \sum_{j=1}^{n-k} \beta_j \mathbf{Aw}_j = \sum_{j=1}^{n-k} \beta_j \mathbf{Aw}_j$$

$\implies \mathbf{y}$ is a linear combination of $\{\mathbf{Aw}_1, \dots, \mathbf{Aw}_{n-k}\}$.

Linear independence: Suppose $\sum_{j=1}^{n-k} \gamma_j \mathbf{Aw}_j = \mathbf{0}$. Then $\mathbf{A}(\sum_{j=1}^{n-k} \gamma_j \mathbf{w}_j) = \mathbf{0} \implies$

$\sum_{j=1}^{n-k} \gamma_j \mathbf{w}_j \in \mathcal{N}(\mathbf{A})$. Since $\{\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}_1, \dots, \mathbf{w}_{n-k}\}$ is a basis of $\mathbb{R}^n$, the vectors $\{\mathbf{w}_1, \dots, \mathbf{w}_{n-k}\}$ are linearly independent and none of them can be expressed as a linear combination of $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$. Therefore, $\sum_{j=1}^{n-k} \gamma_j \mathbf{w}_j = \mathbf{0} \implies \gamma_j = 0$ for all j .

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In conclusion, $\{\mathbf{A}\mathbf{w}_1, \dots, \mathbf{A}\mathbf{w}_{n-k}\}$ is a basis of $\mathcal{C}(\mathbf{A}) \implies \dim(\mathcal{C}(\mathbf{A})) = n - k = n - \dim(\mathcal{N}(\mathbf{A})) \implies \text{rank}(\mathbf{A}) + \dim(\mathcal{N}(\mathbf{A})) = n$.

3.

$$\begin{aligned} \text{rank}(\mathbf{AB}) &\stackrel{1.(a)}{=} \text{rank}(\mathbf{B}^T \mathbf{A}^T) \stackrel{1.(b)}{=} \text{rank}(\mathbf{A}^T) - \dim(\mathcal{C}(\mathbf{A}^T) \cap \mathcal{N}(\mathbf{B}^T)) \\ &\leq \text{rank}(\mathbf{A}^T) \stackrel{1.(a)}{=} \text{rank}(\mathbf{A}) \end{aligned}$$

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Homework 1

Exercise 9: Properties of Eigenvalues and Singular Values

1. Suppose the maximum eigenvalue, minimum eigenvalue and maximum singular value of a given symmetric matrix $\mathbf{A} \in S^n$ are denoted by $\lambda_{\max}(\mathbf{A})$ and $\lambda_{\min}(\mathbf{A})$, respectively. Please show that

$$\lambda_{\max}(\mathbf{A}) = \sup_{\mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq \mathbf{0}} \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}, \quad \lambda_{\min}(\mathbf{A}) = \inf_{\mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq \mathbf{0}} \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}.$$

2. **(Optional)** Suppose $\mathbf{B} = (b_{ij}) \in \mathbb{R}^{m \times n}$ with maximum singular value $\max \sigma_{\max}(\mathbf{B})$.

- (a) Let $\|\mathbf{B}\|_2 := \sup_{\mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{B}\mathbf{x}\|_2}{\|\mathbf{x}\|_2}$. Please show that

$$\sigma_{\max}(\mathbf{B}) = \|\mathbf{B}\|_2.$$

- (b) Please show that

$$\sigma_{\max}(\mathbf{B}) = \sup_{\mathbf{x} \in \mathbb{R}^m, \mathbf{y} \in \mathbb{R}^n, \mathbf{x}, \mathbf{y} \neq \mathbf{0}} \frac{\mathbf{x}^\top \mathbf{B} \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2}.$$

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Solution 9: Properties of Eigenvalues and Singular Values

1. Define Rayleigh quotient:

$$R(x) = \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}$$

Note that $R(\alpha \mathbf{x}) = R(\mathbf{x})$ for any nonzero scalar α . Hence

$$\sup_{\mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq \mathbf{0}} R(x) = \max_{\|\mathbf{x}\|_2=1} \mathbf{x}^\top \mathbf{A} \mathbf{x}, \quad \inf_{\mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq \mathbf{0}} R(x) = \min_{\|\mathbf{x}\|_2=1} \mathbf{x}^\top \mathbf{A} \mathbf{x}$$

Because $\mathbf{x} \mapsto \mathbf{x}^\top \mathbf{A} \mathbf{x}$ is continuous and the unit sphere is compact, both extrema are attained.

By the spectral theorem, there exists an orthogonal matrix $\mathbf{Q}$ and a diagonal matrix $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$ with real eigenvalues λ_i such that $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top$.

Without generality, order the eigenvalues so that $\lambda_{\min} = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n = \lambda_{\max}$.

For any $\mathbf{x} \in \mathbb{R}^n$,

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{x}^\top \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top \mathbf{x} := \mathbf{y}^\top \mathbf{\Lambda} \mathbf{y} = \sum_{i=1}^n \lambda_i y_i^2, \quad \mathbf{x}^\top \mathbf{x} = (\mathbf{Q} \mathbf{y})^\top (\mathbf{Q} \mathbf{y}) = \mathbf{y}^\top \mathbf{y}$$

$$\implies R(x) = \frac{\sum_{i=1}^n \lambda_i y_i^2}{\sum_{i=1}^n y_i^2} \in [\lambda_{\min}, \lambda_{\max}]$$

Let $\mathbf{v}_{\max}$ be a unit eigenvector of $\mathbf{A}$ associated with $\lambda_{\max}$, then

$$R(\mathbf{v}_{\max}) = \frac{\mathbf{v}_{\max}^\top \mathbf{A} \mathbf{v}_{\max}}{\mathbf{v}_{\max}^\top \mathbf{v}_{\max}} = \lambda_{\max}.$$

Similarly, let $\mathbf{v}_{\min}$ be a unit eigenvector associated with $\lambda_{\min}$, then $R(\mathbf{v}_{\min}) = \lambda_{\min}$.

In conclusion,

$$\lambda_{\max}(\mathbf{A}) = \sup_{\mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq \mathbf{0}} \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}, \quad \lambda_{\min}(\mathbf{A}) = \inf_{\mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq \mathbf{0}} \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}.$$

2. (a)

$$\|\mathbf{B}\|_2 = \max_{\|\mathbf{x}\|_2=1} \|\mathbf{B} \mathbf{x}\|_2 = \max_{\|\mathbf{x}\|_2=1} [(\mathbf{B} \mathbf{x})^\top \mathbf{B} \mathbf{x}]^{\frac{1}{2}} = \max_{\|\mathbf{x}\|_2=1} [\mathbf{x}^\top (\mathbf{B}^\top \mathbf{B}) \mathbf{x}]^{\frac{1}{2}}$$

Notice that $\mathbf{B}^\top \mathbf{B}$ is symmetric and positive semi-definite. Without generality, let its eigenvalues be

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0.$$

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and let their corresponding orthogonal normalized eigenvectors be $\mathbf{v}_1, \dots, \mathbf{v}_n \in \mathbb{R}^n$. Then for any $\mathbf{x} \in \mathbb{R}^n$ with $\|\mathbf{x}\|_2 = 1$, there exist scalars $\alpha_1, \dots, \alpha_n$ such that

$$\begin{aligned} \mathbf{x} &= \sum_{i=1}^n \alpha_i \mathbf{v}_i, & \sum_{i=1}^n \alpha_i^2 &= 1 \\ \implies \mathbf{x}^\top (\mathbf{B}^\top \mathbf{B}) \mathbf{x} &= \sum_{i=1}^n \lambda_i \alpha_i^2 \leq \lambda_1 \sum_{i=1}^n \alpha_i^2 = \lambda_1 \end{aligned}$$

Besides, let $\mathbf{x} = \mathbf{v}_1$, then $\mathbf{x}^\top (\mathbf{B}^\top \mathbf{B}) \mathbf{x} = \lambda_1$.

In conclusion, $\|\mathbf{B}\|_2 = \sqrt{\lambda_1} = \sigma_{\max}(\mathbf{B})$.

(b) By Cauchy-Schwarz inequality,

$$\mathbf{x}^\top \mathbf{B} \mathbf{y} \leq \|\mathbf{x}\|_2 \|\mathbf{B} \mathbf{y}\|_2 \leq \|\mathbf{x}\|_2 \|\mathbf{B}\|_2 \|\mathbf{y}\|_2 = \|\mathbf{x}\|_2 \sigma_{\max}(\mathbf{B}) \|\mathbf{y}\|_2$$

Let $\mathbf{x} = \mathbf{u}_1$ and $\mathbf{y} = \mathbf{v}_1$, where $\mathbf{u}_1$ and $\mathbf{v}_1$ are the left and right singular vectors of $\mathbf{B}$ associated with $\sigma_{\max}(\mathbf{B})$. Then

$$\mathbf{x}^\top \mathbf{B} \mathbf{y} = \mathbf{u}_1^\top \mathbf{B} \mathbf{v}_1 = \sigma_{\max}(\mathbf{B}) = \|\mathbf{u}_1\|_2 \sigma_{\max}(\mathbf{B}) \|\mathbf{v}_1\|_2$$

In conclusion,

$$\sigma_{\max}(\mathbf{B}) = \sup_{\mathbf{x} \in \mathbb{R}^m, \mathbf{y} \in \mathbb{R}^n, \mathbf{x}, \mathbf{y} \neq 0} \frac{\mathbf{x}^\top \mathbf{B} \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2}.$$

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Exercise 10: Matrix SVD Decomposition and Pseudoinverse

1. For any real matrix $\mathbf{A} \in \mathbb{R}^{n \times m}$, the **Moore-Penrose generalized inverse** (or pseudoinverse) of $\mathbf{A}$, denoted by $\mathbf{A}^+ \in \mathbb{R}^{m \times n}$, is a matrix that satisfies the following four conditions:

- (a) $\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A}$ (Consistency condition)
- (b) $\mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+$ (Reflexivity condition)
- (c) $(\mathbf{A}\mathbf{A}^+)^\top = \mathbf{A}\mathbf{A}^+$ (Symmetry condition 1)
- (d) $(\mathbf{A}^+\mathbf{A})^\top = \mathbf{A}^+\mathbf{A}$ (Symmetry condition 2)

Suppose that the matrix $\mathbf{A}$ can be decomposed via Singular Value Decomposition (SVD) as $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, Please show that $\mathbf{A}^+ = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^\top$, where $\mathbf{\Sigma}^+ \in \mathbb{R}^{m \times n}$ is defined by:

$$\Sigma_{ij}^+ = \begin{cases} \frac{1}{\Sigma_{ii}} & \text{if } i = j \text{ and } \Sigma_{ii} \neq 0, \\ 0 & \text{otherwise.} \end{cases}$$

2. **(Optional)** Please show that $\mathbf{A}^+$ is unique for any matrix $\mathbf{A} \in \mathbb{R}^{n \times m}$.
3. Consider the linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ where $\mathbf{A} \in \mathbb{R}^{n \times m}$, $\mathbf{x} \in \mathbb{R}^m$, and $\mathbf{b} \in \mathbb{R}^n$. Please show that if the system has no solution (i.e., $\mathbf{b}$ is not in the column space of $\mathbf{A}$), the least squares solution to the system

$$\arg \min_{\mathbf{x} \in \mathbb{R}^m} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2,$$

is given by $\mathbf{x} = \mathbf{A}^+\mathbf{b}$, where $\mathbf{A}^+ \in \mathbb{R}^{m \times n}$ is the Moore-Penrose generalized inverse of matrix $\mathbf{A}$ defined above.

(**Hint:** For any orthogonal matrix $\mathbf{U} \in \mathbb{R}^{n \times n}$ and vector $\mathbf{x} \in \mathbb{R}^n$, then $\|\mathbf{U}\mathbf{x}\|_2 = \|\mathbf{x}\|_2$)

Solution 10: Matrix SVD Decomposition and Pseudoinverse

1. (a)

$$\mathbf{A}\mathbf{A}^+ \mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \mathbf{V}\mathbf{\Sigma}^+ \mathbf{U}^\top \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top = \mathbf{U}\mathbf{\Sigma}\mathbf{\Sigma}^+ \mathbf{\Sigma}\mathbf{V}^\top = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top = \mathbf{A}$$

(b)

$$\mathbf{A}^+ \mathbf{A}\mathbf{A}^+ = \mathbf{V}\mathbf{\Sigma}^+ \mathbf{U}^\top \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \mathbf{V}\mathbf{\Sigma}^+ \mathbf{U}^\top = \mathbf{V}\mathbf{\Sigma}^+ \mathbf{\Sigma}\mathbf{\Sigma}^+ \mathbf{U}^\top = \mathbf{V}\mathbf{\Sigma}^+ \mathbf{U}^\top = \mathbf{A}^+$$

(c)

$$(\mathbf{A}\mathbf{A}^+)^\top = (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \mathbf{V}\mathbf{\Sigma}^+ \mathbf{U}^\top)^\top = \mathbf{U}\mathbf{\Sigma}\mathbf{\Sigma}^+ \mathbf{U}^\top = \mathbf{A}\mathbf{A}^+$$

(d)

$$(\mathbf{A}^+ \mathbf{A})^\top = (\mathbf{V}\mathbf{\Sigma}^+ \mathbf{U}^\top \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top)^\top = \mathbf{V}\mathbf{\Sigma}^+ \mathbf{\Sigma}\mathbf{V}^\top = \mathbf{A}^+ \mathbf{A}$$

2. Assume that there exist two pseudoinverses $\mathbf{B}$ and $\mathbf{C}$ of $\mathbf{A}$. Set $\mathbf{X} = \mathbf{B} - \mathbf{C}$.

Given that $\mathbf{B}$ and $\mathbf{C}$ satisfy the four Moore-Penrose conditions:

$$\mathbf{A}\mathbf{B}\mathbf{A} = \mathbf{A}, \quad \mathbf{B}\mathbf{A}\mathbf{B} = \mathbf{B}, \quad (\mathbf{A}\mathbf{B})^\top = \mathbf{A}\mathbf{B}, \quad (\mathbf{B}\mathbf{A})^\top = \mathbf{B}\mathbf{A}$$

$$\mathbf{A}\mathbf{C}\mathbf{A} = \mathbf{A}, \quad \mathbf{C}\mathbf{A}\mathbf{C} = \mathbf{C}, \quad (\mathbf{A}\mathbf{C})^\top = \mathbf{A}\mathbf{C}, \quad (\mathbf{C}\mathbf{A})^\top = \mathbf{C}\mathbf{A}$$

Then

$$\mathbf{A}\mathbf{X}\mathbf{A} = \mathbf{A}\mathbf{B}\mathbf{A} - \mathbf{A}\mathbf{C}\mathbf{A} = \mathbf{A} - \mathbf{A} = \mathbf{0}$$

$$\mathbf{A}\mathbf{X} = \mathbf{A}\mathbf{B} - \mathbf{A}\mathbf{C} = (\mathbf{A}\mathbf{B})^\top - (\mathbf{A}\mathbf{C})^\top = (\mathbf{A}\mathbf{B} - \mathbf{A}\mathbf{C})^\top = (\mathbf{A}\mathbf{X})^\top$$

$$\mathbf{X}\mathbf{A} = \mathbf{B}\mathbf{A} - \mathbf{C}\mathbf{A} = (\mathbf{B}\mathbf{A})^\top - (\mathbf{C}\mathbf{A})^\top = (\mathbf{B}\mathbf{A} - \mathbf{C}\mathbf{A})^\top = (\mathbf{X}\mathbf{A})^\top$$

$$\Rightarrow \begin{cases} \|\mathbf{A}\mathbf{X}\|_2^2 = \text{tr}((\mathbf{A}\mathbf{X})^\top \mathbf{A}\mathbf{X}) = \text{tr}(\mathbf{A}\mathbf{X}\mathbf{A}\mathbf{X}) = \text{tr}((\mathbf{A}\mathbf{X}\mathbf{A})\mathbf{X}) = \text{tr}(\mathbf{0}\mathbf{X}) = 0 \\ \|\mathbf{X}\mathbf{A}\|_2^2 = \text{tr}((\mathbf{X}\mathbf{A})^\top \mathbf{X}\mathbf{A}) = \text{tr}(\mathbf{X}\mathbf{A}\mathbf{X}\mathbf{A}) = \text{tr}(\mathbf{X}(\mathbf{A}\mathbf{X}\mathbf{A})) = \text{tr}(\mathbf{X}\mathbf{0}) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} \mathbf{A}\mathbf{X} = \mathbf{0} \\ \mathbf{X}\mathbf{A} = \mathbf{0} \end{cases}$$

$$\begin{aligned} \Rightarrow \mathbf{X} &= \mathbf{B} - \mathbf{C} = \mathbf{B}\mathbf{A}\mathbf{B} - \mathbf{C}\mathbf{A}\mathbf{C} \\ &= \mathbf{B}\mathbf{A}\mathbf{B} - \mathbf{C}\mathbf{A}\mathbf{B} + \mathbf{C}\mathbf{A}\mathbf{B} - \mathbf{C}\mathbf{A}\mathbf{C} \\ &= (\mathbf{B} - \mathbf{C})\mathbf{A}\mathbf{B} + \mathbf{C}\mathbf{A}(\mathbf{B} - \mathbf{C}) \\ &= \mathbf{X}\mathbf{A}\mathbf{B} + \mathbf{C}\mathbf{A}\mathbf{X} = \mathbf{0}\mathbf{B} + \mathbf{C}\mathbf{0} = \mathbf{0} \end{aligned}$$

$$\Rightarrow \mathbf{B} = \mathbf{C}$$

In conclusion, the pseudoinverse $\mathbf{A}^+$ is unique.

3. Let $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ be the SVD of $\mathbf{A}$, where $\mathbf{U} \in \mathbb{R}^{n \times n}$ and $\mathbf{V} \in \mathbb{R}^{m \times m}$ are orthogonal matrices, and $\mathbf{\Sigma} \in \mathbb{R}^{n \times m}$ is a diagonal matrix with singular values $\sigma_1, \sigma_2, \dots, \sigma_r$ on the diagonal (where $r = \text{rank}(\mathbf{A})$).

Then

$$\|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2 = \|\mathbf{U}^\top (\mathbf{A}\mathbf{x} - \mathbf{b})\|_2^2 = \|\mathbf{\Sigma}\mathbf{V}^\top \mathbf{x} - \mathbf{U}^\top \mathbf{b}\|_2^2$$

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Let $\mathbf{y} = \mathbf{V}^\top \mathbf{x}$ and $\mathbf{c} = \mathbf{U}^\top \mathbf{b}$. Then the problem reduces to

$$\min_{\mathbf{y} \in \mathbb{R}^m} \|\Sigma \mathbf{y} - \mathbf{c}\|_2^2 = \min_{\mathbf{y} \in \mathbb{R}^m} \sum_{i=1}^r (\sigma_i y_i - c_i)^2 + \sum_{i=r+1}^n c_i^2$$

Note that the second term $\sum_{i=r+1}^n c_i^2$ is constant with respect to $\mathbf{y}$. Therefore, we only need to minimize the first term, which is minimized when $y_i = \frac{c_i}{\sigma_i}$ for $i = 1, 2, \dots, r$ and y_i can be any value for $i = r+1, r+2, \dots, m$.

Hence, the set of least squares minimizers is

$$\mathbf{x} = \mathbf{V} \mathbf{y}, \quad \mathbf{y} = \left(\frac{c_1}{\sigma_1}, \frac{c_2}{\sigma_2}, \dots, \frac{c_r}{\sigma_r}, y_{r+1}, \dots, y_m \right)^\top, y_{r+1}, \dots, y_m \in \mathbb{R}$$

Let $y_i = 0$, $i = r+1, r+2, \dots, m$. Then the least-norm solution is

$$\mathbf{x} = \mathbf{V} \left(\frac{c_1}{\sigma_1}, \frac{c_2}{\sigma_2}, \dots, \frac{c_r}{\sigma_r}, 0, \dots, 0 \right)^\top = \mathbf{V} \Sigma^+ \mathbf{U}^\top \mathbf{b} = \mathbf{A}^+ \mathbf{b}$$

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